Analysis of complicating factors affecting road accidents

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January 2, 2026

Abstract

The objective of this analysis is to examine the influence of complicating factors on road accidents. In particular, we focus on accidents involving complicating factors, such as alcohol, drug influence, poor visibility, weather conditions, and different(unfavorable) road conditions.

1 Analysis of collected statistics

The data include statistics for 2023-2025 collected in the Czech Republic. First, the data are broken down by month and the presence of complicating factors.

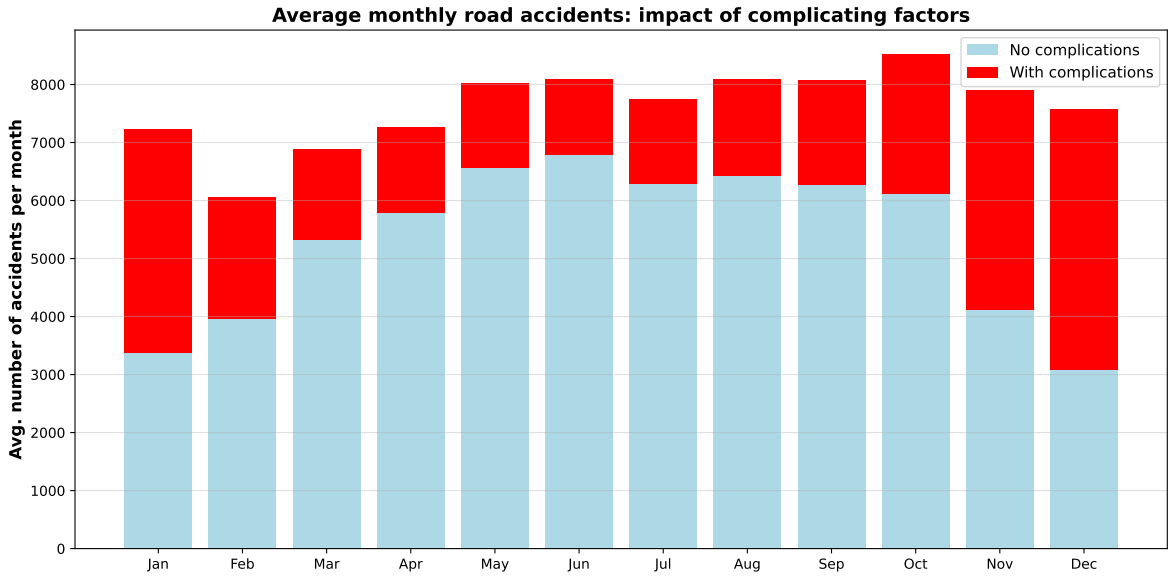


Figure 1: Seasonal distribution of accidents with and without complicating factors

The distribution clearly follows a seasonal pattern, which is further confirmed by aggregated seasonal statistics. For further analysis, it will be more convenient to use text data, and for simplicity, we will collect statistics by season, since based on graph 1, we can clearly see that the distribution corresponds to the seasons, which we will confirm further.

Table 1: Seasonal breakdown of complicating factors

Season	Total	No comp.	Alcohol	Drugs	Visibility	Weather	Road cond.	Comp. rate
Winter	55039	28189	2300	261	8802	7690	22751	48.78%
Spring	66468	53039	3030	293	4643	3150	9316	20.20%
Summer	71804	58513	4125	353	4437	3014	8344	18.51%
Autumn	57173	39351	2501	244	7430	4974	13657	31.17%

2 Further analysis of the impact of complicating factors

Based on the data obtained after processing and systematization, we can perform some calculations according to the table 1.

2.1 Accident volume difference by season

The minimum and maximum total numbers of accidents vary noticeably across seasons.

- The highest number of accidents occurs in Summer (71,804 cases).
- The lowest number of accidents occurs in Winter (55,039 cases).

This corresponds to a 30.46% increase in the number of accidents from winter to summer.

This result suggests that accident frequency is strongly influenced by seasonal traffic intensity, which is typically higher during the summer months due to increased mobility and tourism.

2.2 Complicating factors influence: Winter vs Summer

While summer has the highest absolute number of accidents, winter accidents are significantly more affected by complicating factors.

- In Winter, 48.78% of all accidents involve complicating factors.
- In Summer, only 18.51% of accidents involve complicating factors.

This represents a 163.53% relative increase in the proportion of complicated accidents during winter compared to summer.

Therefore, although fewer accidents occur in winter overall, they are substantially more likely to be associated with adverse conditions.

2.3 Proportions of complicating factors

To better understand the nature of seasonal risks, we analyzed the relative contribution of individual complicating factors among accidents with complications.

Winter

- Alcohol: 8.57%
- Drugs: 0.97%
- Visibility: 32.78%
- Weather: 28.64%
- Road conditions: 84.73%

Summer

- Alcohol: 31.04%
- Drugs: 2.66%
- Visibility: 33.38%
- Weather: 22.68%
- Road conditions: 62.78%

Here we can see that, both in winter and summer, complicating factors in the form of weather conditions are common; however, in winter, this percentage remains higher. We also see an incredible increase in accidents involving alcohol or drugs. Surprisingly, the proportion of accidents caused by visibility problems is almost the same, but they still occur in a third of the cases.

3 Conclusion

Seasonal analysis reveals two fundamentally different accident patterns:

- Summer is characterized by a higher frequency of accidents but relatively fewer complicating factors present, with a stronger influence of behavioral factors such as alcohol and drugs.
- Winter contains fewer accidents overall, but a much higher probability of complications, primarily caused by adverse road and weather conditions.

These findings demonstrate that accident prevention strategies should be season-specific.

- winter safety measures should focus on infrastructure maintenance and weather-related risk mitigation.
- while summer prevention should emphasize behavioral interventions, particularly targeting alcohol-related driving.

4 Note

I tried to make the report fit on one A4 page, but considering that the title, graph, and table take up about 70% of the page, I didn't understand how to achieve this. Considering that it was also necessary to calculate the values and show what these values are for. And, of course, I needed a conclusion summarizing the work done. Therefore, the text was reduced to a minimum without losing its meaning. Some further explanations can be found in the code.